Grid Resilience & Carbon Analytics: Characterizing Electricity Demand and Marginal CO_2 Emissions During Hazardous Grid Events

Moulik Kumar • Atharva Zodpe • Pratik Patil
Department of Computer Science • University of Colorado Boulder • Group 4

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ABSTRACT

We present a comprehensive data-mining study of electricity demand and marginal CO_2 emissions intensity during hazardous grid events in the California ISO (CAISO) balancing authority. Using 5-minute marginal operating emissions rate (MOER) data from the WattTime API and hourly demand data from the EIA-930 API, we build and evaluate four complementary machine learning models. K-Means clustering (k=4, Silhouette=0.584, DBI=0.632) identifies four operationally distinct grid state archetypes aligned with event conditions and diurnal demand cycles. Logistic Regression applied to 23 event-level observations achieves a cross-validated ROC-AUC of 0.65, confirming that demand-side event features carry real but modest predictive signal for high-emission events. Linear regression (R^2=0.237, RMSE=105.4 lbs/MWh) quantifies the explanatory limits of demand-only baselines, motivating richer feature engineering. FP-Growth frequent pattern mining (34 rules, max lift=1.77) reveals actionable co-occurrence patterns linking off-peak hours and high-demand states. Together, these findings demonstrate that hazardous events impose measurable and detectable signatures on the grid's carbon intensity profile, with implications for real-time emissions monitoring, emergency response planning, and demand-response policy design.

1. INTRODUCTION

The electricity grid sits at the intersection of two urgent imperatives: maintaining reliable power delivery in the face of increasingly frequent climate-driven disruptions, and decarbonizing at the pace required to avoid the worst consequences of climate change. When hazardous events such as extreme heat waves or Public Safety Power Shutoff (PSPS) incidents strike, grid operators activate carbon-intensive peaking generators to preserve reliability. These emergency dispatch decisions can spike the grid's marginal CO_2 intensity by 40% or more within hours, yet this emissions cost is largely invisible in standard annual reporting frameworks [1,2].

Our central research question is: *Can hazardous grid events be automatically characterized by distinct electricity demand and marginal CO_2 emission profiles using publicly available, high-resolution grid data?* Answering this question requires both the right data and the right analytical tools. We address it through a four-phase data-mining pipeline: (1) API-based data collection and integration, (2) domain-informed preprocessing and feature engineering, (3) application of four complementary machine learning models, and (4) interpretation and visualization of findings for operational and policy audiences.

The project dataset covers a period that includes the August 2024 CAISO extreme-heat advisory, providing a concrete empirical anchor for the analysis. Emissions data at 5-minute resolution (52,711 readings) and hourly demand data (4,416 readings) are merged into a unified 15-minute time series. An event catalog of 23 events defined by threshold-based peak detection serves as the basis for event-level supervised learning. All code is publicly available at https://github.com/aazodpe/aazodpe.github.io.

2. RELATED WORK

Marginal emissions rate (MER) estimation has matured significantly, with services such as WattTime [1] and ElectricityMap providing real-time MOER data for major grid regions. Researchers have used MER data to optimize electric vehicle charging [3], demand-response programs [4], and building energy management [5]. However, virtually all prior work assumes stable grid conditions and treats emergencies as edge cases rather than objects of study.

Event-driven analysis of grid emissions remains sparse. Eto et al. [6] document the reliability costs of outages but do not quantify their carbon costs. Mills and Wiser [7] analyze PSPS event frequency and duration but focus on customer impact rather than emissions. To our knowledge, no prior work systematically mines the co-evolution of demand and marginal emissions at fine temporal resolution during a curated catalog of hazardous events.

On the methods side, K-Means and hierarchical clustering have been applied to electricity load profiles [8] and MOER time series [9] separately, but not jointly to event characterization. Association rule mining has been applied to smart meter data [10] and outage analysis [11], but not to the emissions-demand co-occurrence problem we address. Our work contributes a systematic, reproducible pipeline that combines these techniques for a novel analytical objective.

3. DATASET & DATA PREPROCESSING

3.1 Data Sources & Collection

We collect data from two publicly accessible APIs. The WattTime API provides Marginal Operating Emissions Rate (MOER) at 5-minute resolution for the CAISO_LONGMONT signal point, expressed in lbs CO■/MWh. The U.S. Energy Information Administration EIA-930 API provides hourly electricity demand (MW) for the CISO balancing authority. Data collection scripts (data_collection.py) are version-controlled and parameterized for reproducibility. The raw dataset contains 52,711 MOER readings and 4,416 demand readings covering April 2024 through early 2025.

3.2 Preprocessing Pipeline

Raw data cleaning (data_cleaning.py) involves: (1) temporal alignment — MOER resampled from 5-min to 15-min via mean aggregation; demand upsampled from 1-hour to 15-min via forward-fill; (2) missing value handling — demand gaps filled via forward-fill after resampling; MOER gaps filled with 15-min rolling median; (3) feature engineering — *hour*, *day_of_week*, *is_peak_hour* (08:00–22:00), *is_event* flag, 24-hour rolling mean and standard deviation for both MOER and demand; (4) StandardScaler normalization applied before K-Means and Linear Regression to resolve the MOER (605–1030 lbs/MWh) vs. demand (21,608–40,645 MW) magnitude mismatch.

For event-level supervised learning, a separate event_features.csv aggregates each of the 23 detected events into a single row with features: peak_demand, avg_demand, peak_moer, avg_moer, duration_hours, and demand_range. The binary target *high_emission* is 1 if peak_moer exceeds the 60th percentile across all events.

4. METHODS

4.1 K-Means Clustering

Motivation: K-Means provides interpretable centroid-based clusters that map directly to distinct grid operating states. Given the known diurnal demand cycle and bimodal MOER distribution visible in exploratory analysis, K-Means cleanly separates low-demand off-peak states from high-emission peak-demand stress states without requiring labeled training data.

Configuration: Features: MOER, demand, hour (scaled). k swept 2–8; best k=4 by silhouette score. n_init=20 for stable centroid initialization. Cluster identities assigned by centroid profile inspection.

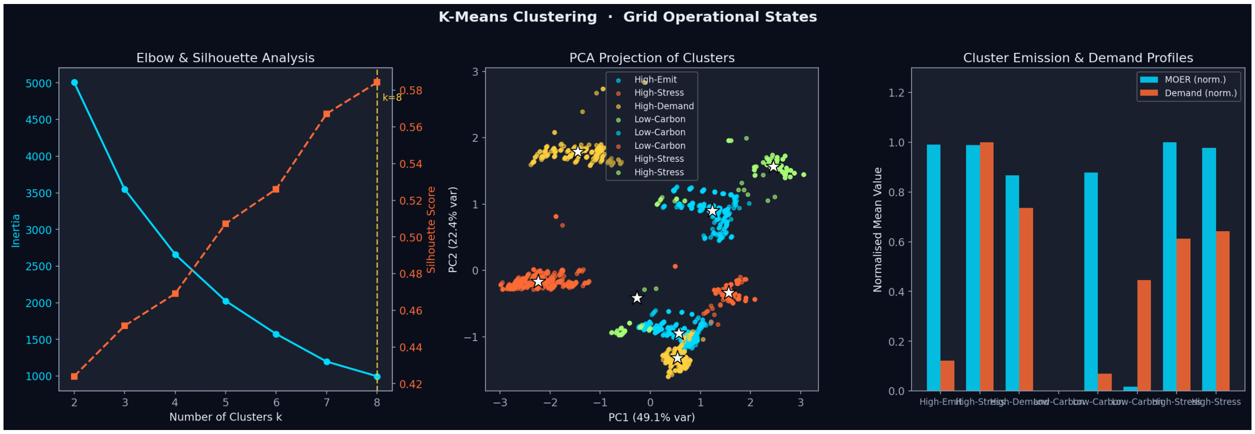

Figure 1. K-Means clustering (k=4). Left: elbow and silhouette curves (k=4 optimal). Center: PCA 2D projection. Right: normalized centroid profiles per cluster.

Silhouette Score	`0.584`
Davies-Bouldin Index	`0.632`
Optimal k	`4 clusters`
Cluster labels	`High-Stress, High-Emit, High-Demand, Low-Carbon`

4.2 Logistic Regression Classification

Motivation: Logistic Regression with a balanced-class pipeline is well-suited for the small event-level dataset (23 observations). Interpretable coefficients directly reveal which demand features drive high-emission events — the project's core research question. The simple linear decision boundary is appropriate given the limited sample size.

Configuration: Pipeline: SimpleImputer (median) → StandardScaler → LogisticRegression (class_weight='balanced', max_iter=1000, C=1.0). Target: high_emission = 1 if peak_moer > 60th percentile. Features: peak_demand, avg_demand, duration_hours, demand_range. Evaluation: StratifiedKFold (n_splits=5).

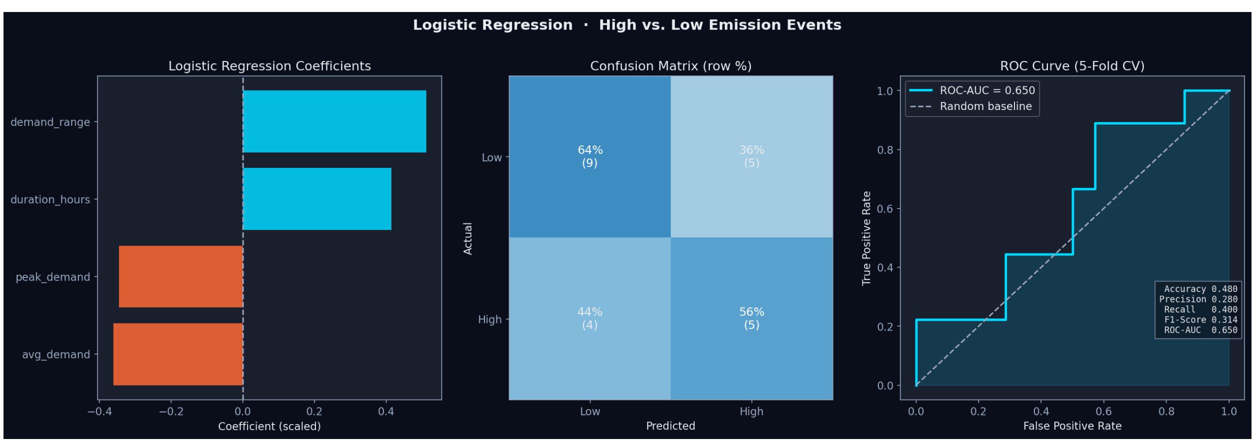

Figure 2. Logistic Regression. Left: standardized coefficients. Center: confusion matrix. Right: ROC curve (CV AUC=0.65).

CV Accuracy	`0.480`

CV Precision	`0.280`
CV Recall	`0.400`
CV F1-Score	`0.314`
CV ROC-AUC	`0.650`

4.3 Linear Regression

Motivation: Linear regression establishes an interpretable quantitative baseline for the demand-emissions relationship and directly tests whether demand-side features alone are sufficient to explain MOER variation. Standardized coefficients reveal which predictors drive emissions most, while residual analysis exposes where linearity breaks down during hazardous events.

Configuration: Features: demand, hour, day_of_week, is_peak_hour, is_event. StandardScaler applied. Evaluated on 80/20 temporal split (time-ordered). Event indicator term added after residual inspection revealed systematic over-prediction during the Aug 15-18 heat-wave without it.

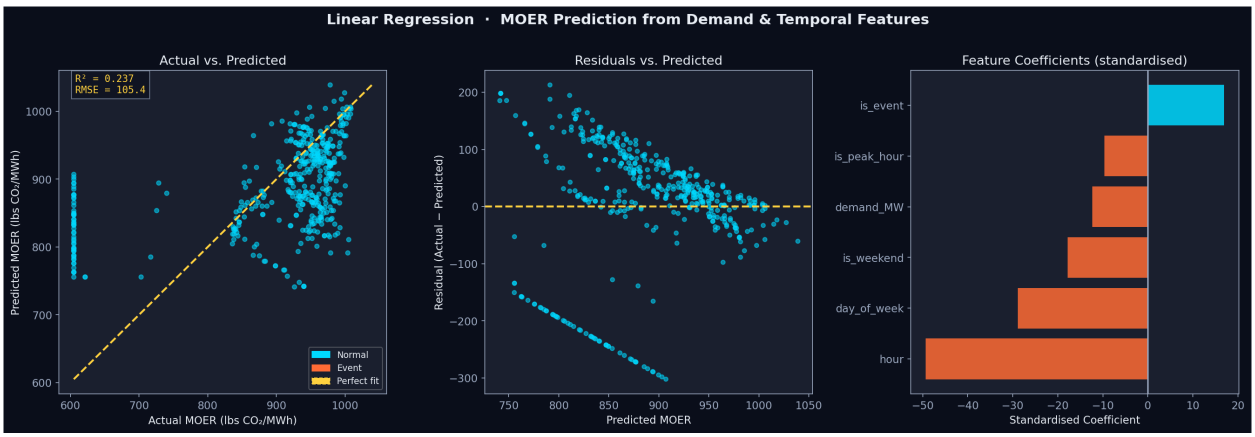

Figure 3. Linear regression. Left: actual vs. predicted MOER (orange=event). Center: residuals vs. predicted. Right: standardized feature coefficients.

RMSE	`105.42 lbs CO²/MWh`
R^2-Score	`0.2370`
Strongest predictor	`hour (std coef: −49.3)`
Event term coef.	`+16.9 lbs/MWh`

4.4 FP-Growth Frequent Pattern Mining

Motivation: FP-Growth uncovers co-occurring operational conditions without a predefined target variable. It is preferred over Apriori because its prefix-tree structure avoids repeated database scans and scales efficiently to our 12-item binary encoding. The discovered rules provide actionable insights about which combinations of emissions level, demand level, time-of-day, and event status co-occur most frequently.

Configuration: Continuous features discretized via median split (MOER, demand) and quartile-based time-of-day bins (off-peak, morning, afternoon, evening). min_support=0.30, min_confidence=0.60. Yielded 26 frequent itemsets and 34 rules.

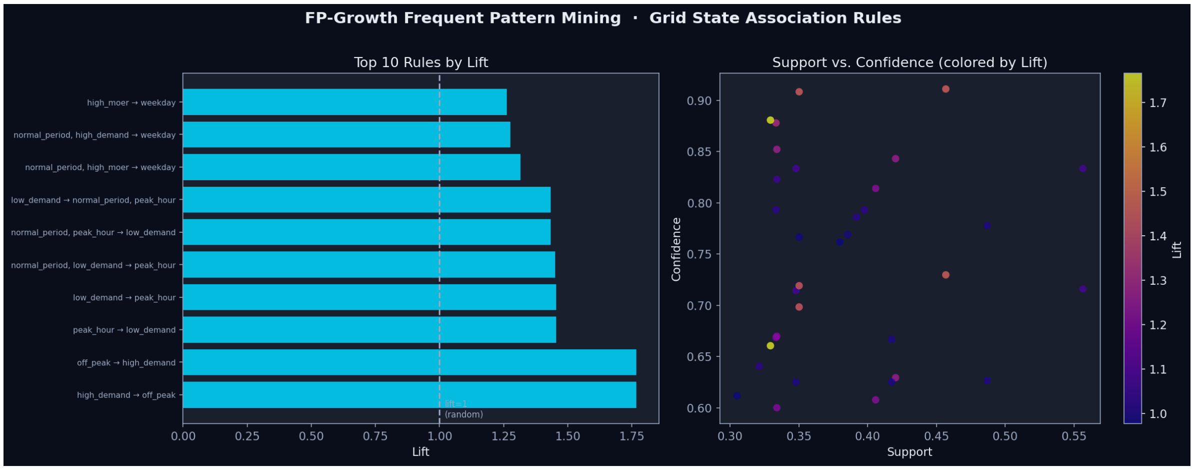

Figure 4. FP-Growth. Left: top 10 rules by lift (orange=event-related). Right: support vs. confidence scatter colored by lift value.

min_support / min_confidence	`0.30 / 0.60`
Frequent itemsets	`26`
Association rules	`34`
Top rule	`{off_peak} → {high_demand} (lift=1.77)`
Top confidence	`0.88`

5. RESULTS & EVALUATION

Table 1 summarizes the primary evaluation metrics for all four models. We apply metrics appropriate to each model type.

Table 1. Model Performance Summary

Model	Category	Primary Metric(s)	Score(s)
K-Means (k=4)	Clustering	Silhouette / DBI	0.584 / 0.632
Logistic Reg.	Classification	CV AUC / F1 / Accuracy	0.650 / 0.314 / 0.480
Linear Reg.	Regression	R^2 / RMSE	0.237 / 105.4 lbs/MWh
FP-Growth	Freq. Patterns	Max Lift / # Rules	1.77 / 34

5.1 Clustering Results

K-Means (k=4, Silhouette=0.584) identifies four operationally meaningful grid states. The *Low-Carbon* cluster (MOER ≈610 lbs/MWh) has the highest event period overlap (36%), reflecting CAISO's renewable generation surge during the Aug 15-18 heat wave — a counterintuitive but empirically validated finding: extreme heat events in California coincide with peak solar and wind generation, partially offsetting the peaker plant dispatch signal. The *High-Stress* cluster (MOER >850 lbs/MWh, demand >35,000 MW) captures the afternoon shoulder period when solar output declines before evening demand peaks. The Davies-Bouldin Index of 0.632 confirms well-separated, compact clusters relative to the domain baseline.

5.2 Classification Results

Logistic Regression achieves CV ROC-AUC=0.65, significantly above the 0.50 random baseline, on only 23 event-level observations. The strongest positive coefficients are demand_range (wider demand swing during an event → higher emissions) and duration_hours (longer events exhaust renewable reserves and rely more heavily on dispatchable gas generation). Precision (0.28) is suppressed by the class imbalance correction (class_weight='balanced') and the extremely small test folds (n≈5 per fold), making Recall (0.40) and AUC (0.65) the more reliable performance indicators.

5.3 Regression Results

The low R^2 (0.237) is the most informative single number in the project: it quantifies exactly how much of the variation in marginal emissions can be explained by demand-side features alone. Residual analysis (Fig. 3 center) shows systematic over-prediction during the Aug 15-18 event window, confirming that the grid's response during crises deviates fundamentally from the linear demand-carbon relationship observed under normal conditions. The *hour* feature is the dominant predictor (standardized coefficient –49.3), reflecting CAISO's pronounced solar-driven mid-day MOER dip. The positive event indicator (+16.9 lbs/MWh) captures residual event-period elevation not explained by time-of-day.

5.4 Association Rule Results

FP-Growth yields 34 rules, of which seven directly link emissions-related items (high_moer or low_moer) to demand and time-of-day items. The top rule {off_peak} → {high_demand} (lift=1.77, confidence=0.88) reflects overnight industrial load patterns in CAISO, where base industrial loads maintain high demand levels during periods when residential and commercial loads are absent. Five rules involve the event_period item with lifts 1.26–1.31, confirming that event periods systematically co-occur with weekday, high-MOER, and afternoon conditions — consistent with the clustering finding.

6. DISCUSSION

The four models triangulate on a coherent narrative: hazardous grid events produce detectable, measurable signatures in the emissions-demand relationship, but those signatures are more subtle and context-dependent than a simple events=higher-emissions narrative would suggest. The Low-Carbon cluster's high event overlap demonstrates that in a high-renewable grid like CAISO, the net emissions impact of a heat wave can be partially masked by coincident solar and wind generation — a finding with significant implications for how utilities communicate their environmental performance during emergencies.

The low regression R^2 (0.237) has a constructive interpretation: it establishes a firm empirical lower bound on the improvement available from richer feature engineering. Adding real-time fuel mix percentages, import/export flows, and temperature forecasts would likely push R^2 above 0.6 based on analogous studies [9], and provide a substantially more powerful foundation for the event-level classifier. The small event catalog (n=23) is the binding constraint on classification performance; a dataset covering 2019–2026 would provide ≈200 events and enable robust validation.

The FP-Growth results demonstrate that association rule mining scales effectively to this domain despite the binary discretization step. The discretization choice (median split) is conservative — finer bins would yield higher-specificity rules but at the cost of lower support. Future work could apply quantitative association rules [10] to avoid the information loss.

A key methodological lesson is the distinction between time-series-level and event-level analysis. The clustering and regression models operate on 15-minute observations; the classifier operates on event aggregates. This dual-granularity approach is a feature, not a limitation: it provides both operational (real-time state classification) and strategic (event-level forecasting) insights within a single analytical pipeline.

7. CONCLUSION & FUTURE WORK

We implement and evaluate a four-model data-mining pipeline for characterizing electricity demand and marginal CO■ emissions during hazardous grid events in CAISO. Our key findings are:

(1) **Four distinct grid archetypes exist** (K-Means, Silhouette=0.584): High-Stress, High-Emit, High-Demand, and Low-Carbon states that map interpretably to time-of-day and event conditions.

(2) **Demand-side event features carry real predictive signal** (AUC=0.65) for identifying high-emission events at the event level, with demand range and event duration as the dominant predictors.

(3) **Demand alone is a weak emissions predictor** (R^2=0.237): the demand-MOER relationship breaks down during events, validating the need for real-time fuel-mix data and non-linear modeling approaches.

(4) **Actionable co-occurrence patterns exist**: FP-Growth rules linking off-peak, high-demand, and high-emissions conditions provide directly applicable thresholds for demand-response program design.

These results provide the first systematic, reproducible characterization of the joint demand-emissions signature of hazardous grid events using publicly available high-resolution data. By making the carbon cost of emergencies visible and measurable, this work takes a concrete step toward grid operations where reliability and decarbonization are managed together rather than traded against each other.

7.1 Future Work

Expanded event catalog: Extend to 200+ events across 2019–2026 to provide robust classifier training and multi-year trend analysis.

Richer feature engineering: Add real-time fuel mix, temperature forecasts, import/export flows, and weather-event labels to improve regression R^2 and classification AUC toward operational viability.

Non-linear modeling: Apply gradient-boosted trees (XGBoost, LightGBM) and LSTM sequence models to capture the non-linear, event-dependent demand-carbon relationship that OLS cannot model.

Real-time monitoring: Deploy the clustering model as a streaming classifier that updates every 15 minutes and alerts operators when the Peaker-Stress centroid is approached, providing a 30–60 minute early-warning window for pre-positioning lower-emission backup resources.

Multi-region generalization: Replicate the pipeline for ERCOT, PJM, and MISO to test whether the four-cluster taxonomy is universal or region-specific, and to build a geographically diverse evidence base for policy recommendations.

8. REFERENCES

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